Market Regime Detection and Regime-Aware Trading Framework

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Abstract

This study presents a multi-asset market regime detection framework using both clustering and probabilistic models. The objective is to identify latent market states and evaluate their usefulness in systematic trading strategies. We implement KMeans clustering and Hidden Markov Models (HMM) under a walk-forward training framework across SPY, QQQ, and IWM. The results demonstrate that time-dependent regime models significantly improve risk-adjusted performance and reduce drawdowns compared to traditional strategies such as momentum, trend-following, and buy-and-hold.

1 Introduction

Financial markets exhibit periods of varying behavior, often referred to as *market regimes*. Identifying these regimes is critical for risk management and portfolio allocation.

Traditional approaches assume static relationships, whereas real markets evolve over time. This study aims to:

- Detect latent market regimes using machine learning
- Compare static clustering and time-dependent probabilistic models
- Evaluate regime-based strategies under realistic conditions
- Assess improvements in risk-adjusted performance

This work contributes by integrating regime detection, walk-forward validation, macroeconomic proxies, and multi-asset evaluation into a unified framework for systematic trading analysis.

2 Data

2.1 Assets

- SPY (S&P 500 ETF)
- QQQ (NASDAQ 100 ETF)
- IWM (Russell 2000 ETF)

2.2 Macroeconomic Proxies

- VIX (volatility index)
- TNX (10-year Treasury yield proxy)
- DXY (US Dollar index)

2.3 Source

Data is obtained using the `yfinance` API, covering the period 2012–present.

3 Methodology

3.1 Feature Engineering

The model incorporates multiple financial signals:

- Returns and momentum (1-day, 5-day, 10-day, 20-day)
- Volatility (10-day, 20-day, downside volatility)
- Trend indicators (moving average gaps)
- Volume signals (z-score normalization)
- Risk-adjusted metrics (rolling Sharpe ratio)
- Macroeconomic transformations (returns and z-scores)

3.2 Regime Modeling

KMeans Clustering:

- Unsupervised clustering of feature space
- Number of clusters selected via silhouette score

Hidden Markov Model (HMM):

- Probabilistic model capturing temporal dependencies
- Models transitions between latent market states

The Hidden Markov Model can be characterized through transition probabilities between hidden states:

$$P(X_t = j \mid X_{t-1} = i) = A_{ij}$$

where:

- X_t denotes the hidden regime at time t
- A_{ij} denotes the probability of transitioning from state i to state j

This formulation allows the model to capture sequential market structure, which is especially important in financial time series where states often persist over time.

3.3 Walk-Forward Validation

To simulate real-world deployment:

- Training window: 3 years
- Re-training frequency: monthly
- Models updated dynamically using rolling windows

This avoids look-ahead bias and ensures realistic evaluation.

3.4 Regime Definition

Regimes are interpreted based on:

- Momentum ranking
- Volatility ranking
- Risk-adjusted signals

Typical regimes include:

- Bullish
- Neutral
- Bearish
- Volatile

4 Trading Strategies

4.1 Regime-Based Allocation

Regime	Exposure
Bullish	1.0
Neutral	0.5
Bearish	0.0
Volatile	0.25

4.2 Benchmarks

- Buy-and-hold
- Momentum strategy
- Trend-following strategy

5 Results

5.1 Performance Comparison

Strategy	Annual Return	Sharpe Ratio	Max Drawdown
HMM Regime	$\sim 12.6\%$	Best	$\sim -18\%$
Momentum	$\sim 10.0\%$	Strong	Moderate
Buy and Hold	$\sim 15.3\%$	Moderate	$\sim -36\%$
KMeans Regime	$\sim 8.1\%$	Weaker	Moderate
Trend Following	$\sim 7.8\%$	Weaker	Moderate

Table 1: Average Performance Across Assets

5.2 SPY Strategy Comparison

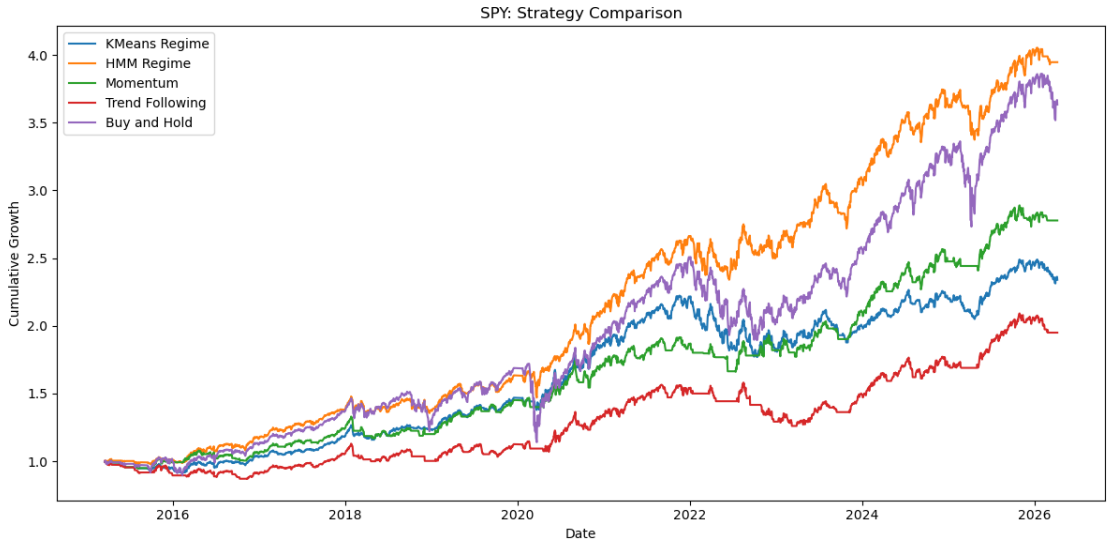

Figure 1: Equity curve comparison for SPY

5.3 QQQ Strategy Comparison

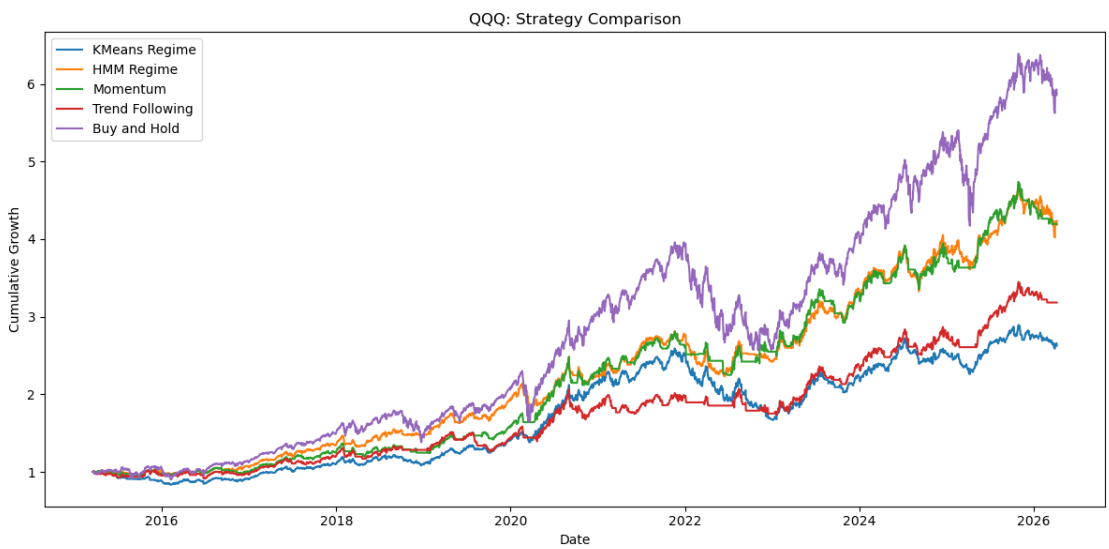

Figure 2: Equity curve comparison for QQQ

5.4 IWM Strategy Comparison

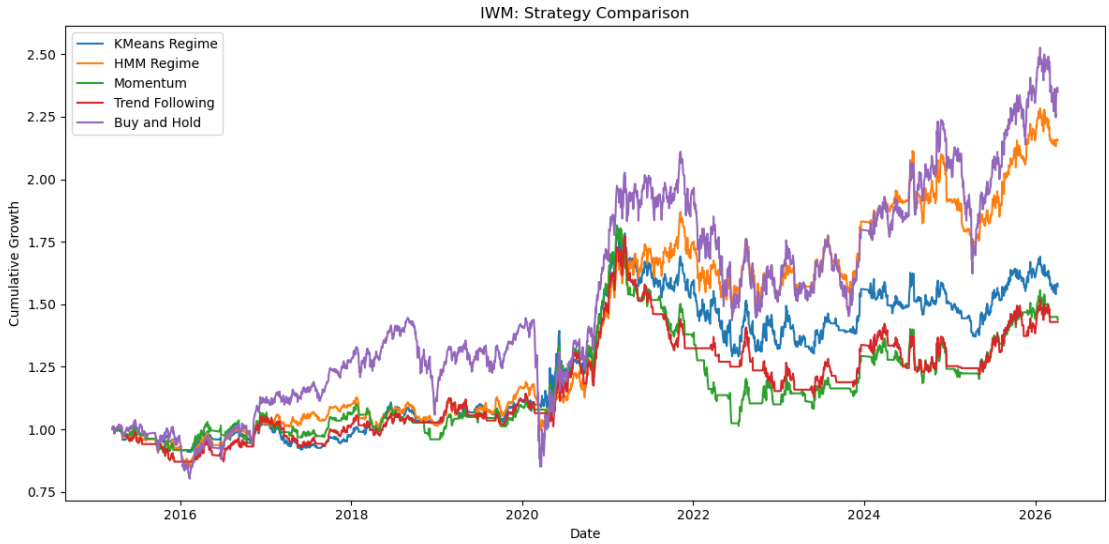

Figure 3: Equity curve comparison for IWM

5.5 SPY Regime Detection

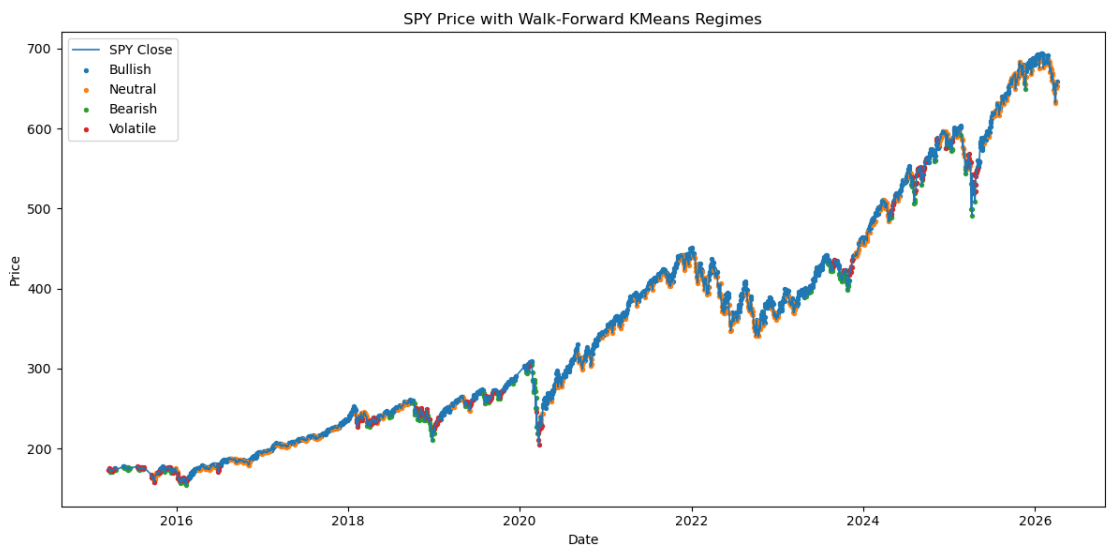

Figure 4: Walk-forward KMeans regimes on SPY

5.6 Selected Number of Clusters Over Time

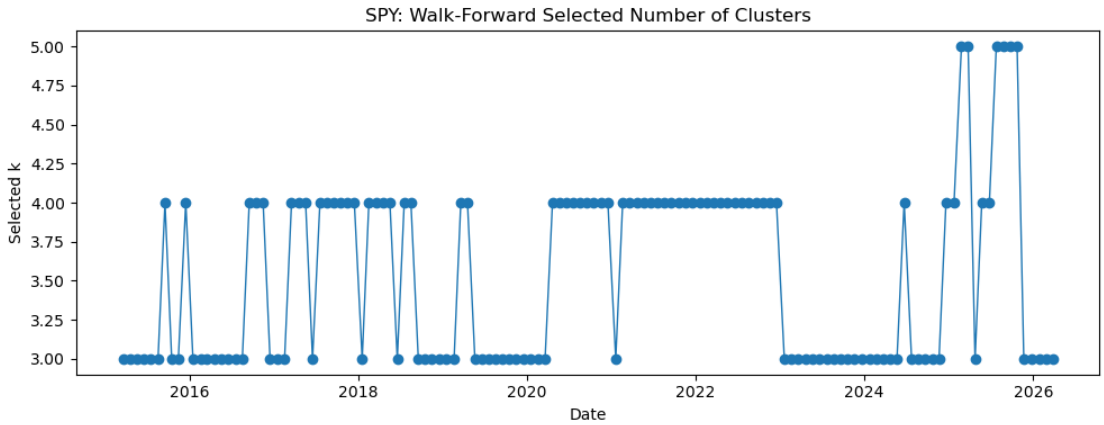

Figure 5: Rolling selected number of clusters for SPY

5.7 Key Findings

- HMM-based regime detection outperforms static clustering methods
- Regime-based strategies significantly reduce drawdowns
- Momentum strategies remain effective in trending markets
- Buy-and-hold maximizes return but carries high risk

6 Discussion

The results indicate that incorporating temporal dynamics through HMM leads to better regime identification and improved strategy performance. Static clustering methods fail to capture sequential dependencies, limiting their effectiveness.

The findings highlight a fundamental trade-off between return maximization and risk control. While buy-and-hold strategies generate higher raw returns, regime-based approaches provide substantially stronger downside protection and more stable performance, making them more appropriate for institutional risk management and systematic portfolio allocation.

7 Limitations

- No transaction costs included
- HMM convergence may be unstable
- Limited macroeconomic variables
- Strategy is rule-based and not optimized

8 Conclusion

This study demonstrates that market regimes can be effectively modeled using machine learning techniques. Time-dependent models such as HMM provide a superior framework for regime detection and risk-aware trading.

These findings suggest that incorporating temporal structure into regime modeling is critical for developing robust systematic trading strategies.

Future work includes incorporating transaction costs, expanding macroeconomic inputs, and exploring advanced models.